

KALEB BOYD

Buford, Georgia

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EXPERIENCE

GRVTY Corp.

Geospatial Engineer

January 2025 - Present

Remote

- Developed a target and custody fusion engine stack processing thousands of observations per second from multiple disparate intelligence sources across classified networks, providing analysts an automated and scalable solution for Air Force 480th ISR Wing, Air Force Combat Command, and Air Force Research Laboratory.
- Built distributed tracking system utilizing advanced filtering and data association techniques including particle/Kalman filters, Joint Probabilistic Data Association, and multiple-hypothesis tracking for simultaneous multi-target tracking in near real-time.
- Implemented scalable tech stack using Kafka for data ingestion, FastAPI for API services, and Kubernetes for deployment management, ensuring fault tolerance and high availability.
- Delivered contextual and spatio-temporal correlations to users through real-time processing capabilities, enhancing intelligence analysis workflows.

Secured Transportation Services

Data Scientist

October 2024 - January 2025

Buford, GA

- Developing a custom in-house Large Language Model (LLM) tailored for company needs.
- Utilizing local LLMs fine-tuned with Low-Rank Adaptation (LoRA) and In-Context Learning techniques, including Retrieval-Augmented Generation (RAG).
- Enhancing company capabilities by leveraging generative AI to improve deliverable speed and accuracy.
- Implemented automation engineering solutions, reducing processing time of certain tasks from 5 hours to minutes.

United States Navy

Cyber Threat Intelligence Analyst

May 2023 - September 2024

Pensacola, FL

- Served as a Cyber Threat Intelligence Analyst at Naval Information Operations Command - Pensacola, responsible for implementing defensive and offensive cyberspace mission tasking.
- Collected and processed intelligence to generate products satisfying cyberspace intelligence requirements and essential elements of information.
- Developed target packages supporting cyberspace operations and disseminated cyber intelligence information through briefings, written reports, and messages.
- Deployed to Key Infrastructure and Critical Resource DODIN sites in response to intelligence on potential Advanced Persistent Threat (APT) cyber activity, providing targeted defense and safeguarding these critical networks.
- Utilized the HELK stack (Hunting, Elasticsearch, Logstash, Kibana) to identify and remediate potential malicious cyber activity on numerous Department of Defense (DoD) Networks.
- Applied advanced Artificial Intelligence techniques, including Natural Language Processing and Machine Learning, to enhance network traffic anomaly detection and strengthen cybersecurity measures within the Department of Defense. These efforts led to a 90% reduction in processing time, significantly improving defense against malicious actors in the cyber domain.

- Attached to Fleet Intelligence Detachment, D.C. (FID DC) as a Geospatial Intelligence Analyst.
- Utilized imagery analysis to create enhanced imagery products adhering to National Geospatial Intelligence (NGA) standards through analyzing overhead and aerial imagery developed by photographic and electronic means.
- Deployed aboard the USS CARL VINSON CVN-70 (July 2021 - Early 2022) supporting Carrier Strike Group One (CSG-1) and multi-national operations afloat.
- Developed data analysis tools using Python that provide a faster and more robust way to filter, organize, and display data collected daily via overhead sensors.
- Utilized QGIS software with Python to display geospatial trend analysis of the maritime and air domain for the area of responsibility of CSG-1 resulting in an unprecedented level of awareness that is now being adopted fleet-wide.
- Created and maintained an SQL database fusing data from four on-board intelligence watch floors to establish better cohesion and communication of data across the intelligence enterprise.
- Assisted in the development of data engineering and data analysis tool sets for the Navy’s Project Overmatch program.
- Served as the first tactical afloat data scientist, pioneering the integration of data science on deployment to enhance decision-making and operational productivity. This role catalyzed the ‘Data Science at Sea’ initiative, now being adopted fleet-wide to drive data-informed strategies across naval operations.
- Contributed to the Data Science Working Group for Naval Information Forces, advocating for improved data literacy and specialized training to support tactical data scientists in operational settings.
- Served as departmental Leading Petty Officer for Training directly responsible for managing a team of 7 sailors who provided qualification and pre-deployment training to over 100 sailors.

CERTIFICATIONS

CompTIA	Cybersecurity Analyst (CySA+)
National Geospatial-Intelligence Agency	GEOINT Professional Certification - Fundamental, Imagery Analysis
Department of Defense	Intelligence Fundamentals Professional Certification
Clearance	Current TS/SCI (with Counterintelligence Polygraph)

TECHNICAL STRENGTHS

Computer Languages	Python, R, JavaScript, Bash, Rust
Data Analysis / Visualization	Pandas, Numpy, Matplotlib, Plotly, Dash, Seaborn, ggplot, D3, QGIS
Data Engineering	MySQL, PostgreSQL, Redis, Databricks, Logstash
Data Streaming / Search	Kafka, Kibana, Elasticsearch
Machine Learning Frameworks	Tensorflow, Keras, Scikit-learn, PyTorch
Signal Processing / Advanced Analytics	Particle Filters, Kalman Filters, JPDA, Multi-hypothesis Tracking
Cloud / DevOps	Kubernetes, Docker, Docker Compose
Back-end Web Development	Django, Flask, FastAPI, Actix-Web
Front-end Web Development	HTML, CSS, React, Vue

EDUCATION

Georgia Institute of Technology	August 2024
Master of Science: Analytics	
Focus in Computational Data Analytics (Reinforcement Learning, High-Dimensional Data Analysis)	
University of Georgia	May 2018
Bachelor of Science: Civil Engineering	